

THE CURRENT REALITY

Every alarm says *“all clear”*

Revenue is dropping. Users are churning. The system is failing.
And every single dashboard shows green.

\$2.4M

lost in 72 hours

11 hrs

to find root cause

0

alerts fired

Diagnosis: Six fatal flaws in legacy monitoring

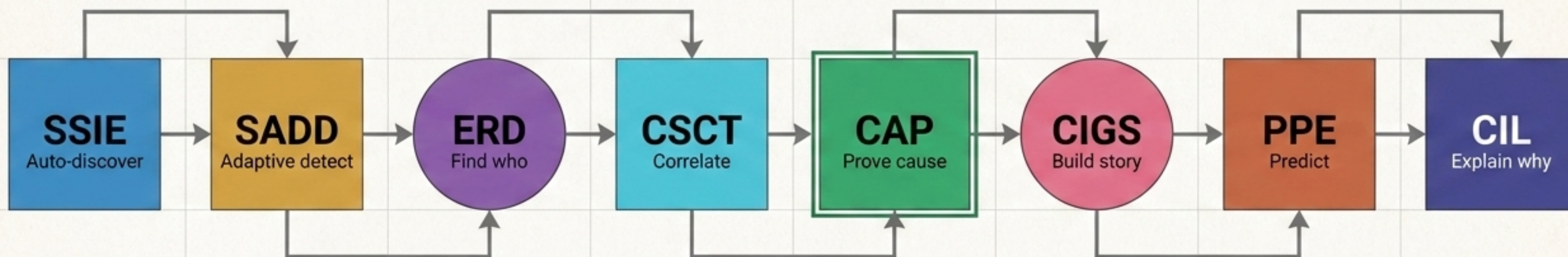
01	Blind to shape changes	Watches single numbers against fixed thresholds. Misses distribution shifts entirely.
02	One method for everything	Uses the same threshold logic for spikes, drifts, and shifts. Wrong tool for 2 of 3.
03	Can't see across layers	Raw events, features, and model outputs live in separate tools. No connection.
04	Correlation, not causation	These might be related. That's not proof. That's a guess.
05	No memory	Knowledge lives in one engineer's head.
06	Always reactive	Alerts fire after impact. The damage is already done.

What if the system could see between the layers?

Correlation is a guess. Thresholds are blind. Six flaws require six novel solutions. We must stop watch numbers against lines, and start watching the **geometric shape** and **proven causal links** between system layers.

Eight modules. One pipeline.

Each solving a problem no other system can.

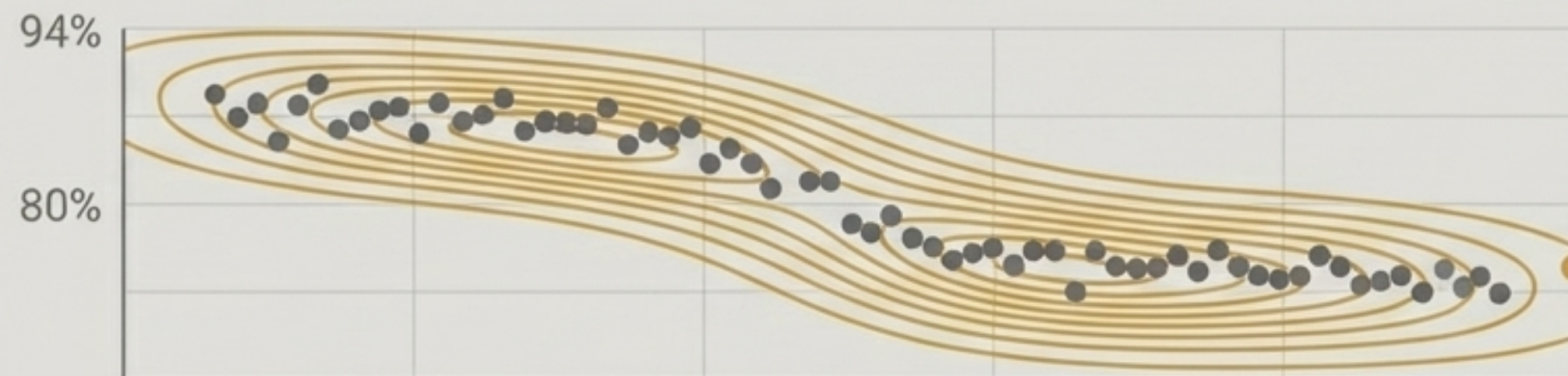


Phase 1: Detection — SADD

The Old Way



The SADD Way



$$S_0: \frac{|\text{value} - \text{median}|}{\text{MAD}}$$

$$S_1: \sum (\text{curr}\% - \text{base}\%) \times \ln(\text{curr}\% / \text{base}\%)$$

$$S_2: \frac{\text{count}(\text{outside 95\% interval})}{\text{total}}$$

Three adaptive engines watch the SHAPE of the entire distribution. MAD Z-scores for spikes. PSI histograms for shifts.

Catches anomalies that are invisible to threshold monitoring.

Phase 1.5: Isolation — ERD's Unfair Advantage



99.96%
Search Space Pruned

Information-theoretic beam search balances the INFORMATION gained versus the COST to describe the group.

When cost exceeds gain, it stops looking.

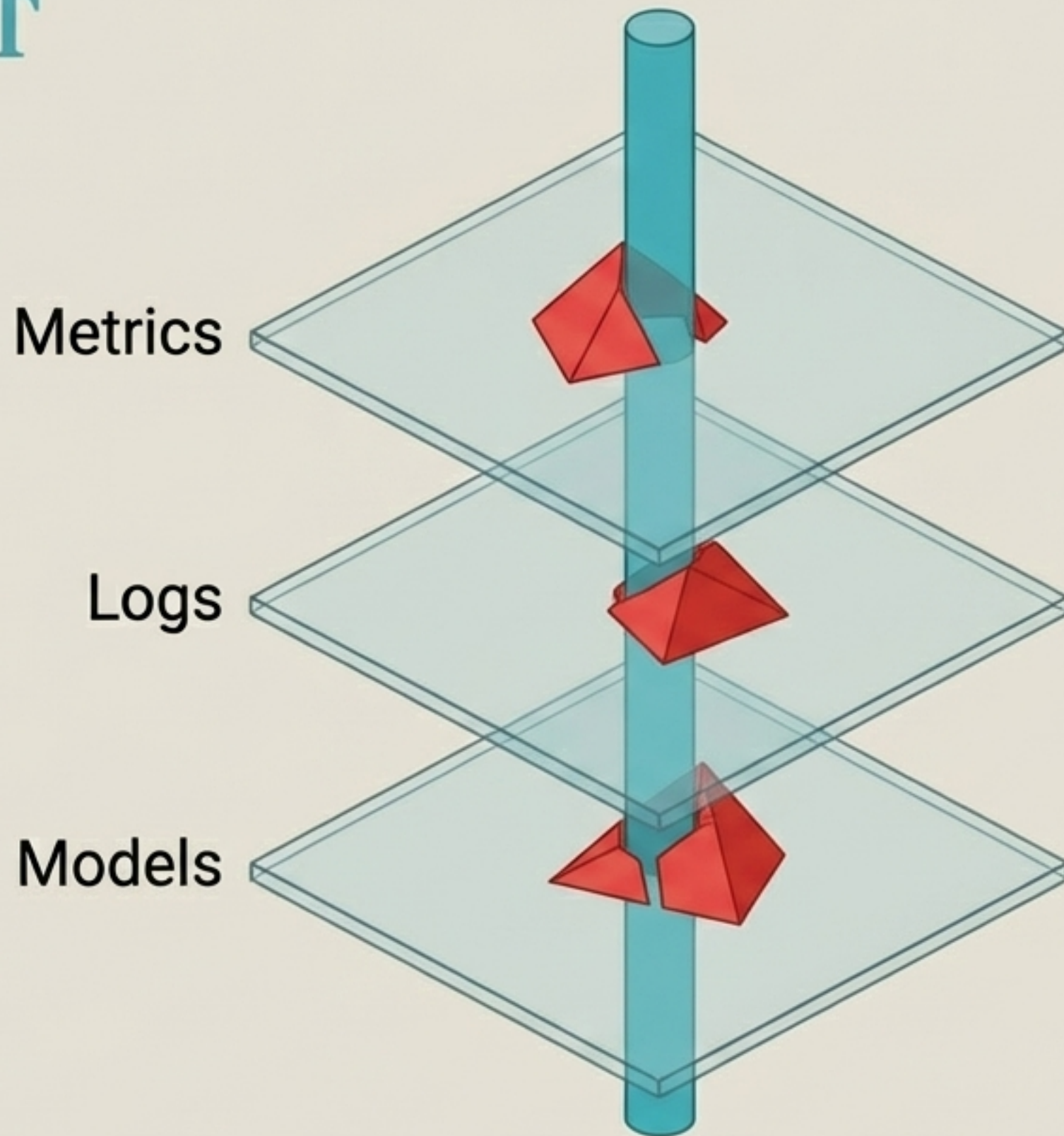
Checking 120,000 groups takes hours; ERD finds the exact subgroup in seconds.

$$\begin{aligned} \text{Score} = & \log_2(\text{concentration}) \times \text{entities} \\ & - \text{depth} \times \log_2(\text{cardinality}) \end{aligned}$$

Phase 2: Propagation — CSCT

The Four-axis cross-stratum tensor computationally connects seemingly unrelated events across different tools using four independent dimensions: entity overlap, learned lag, hierarchy matching, and topic embeddings.

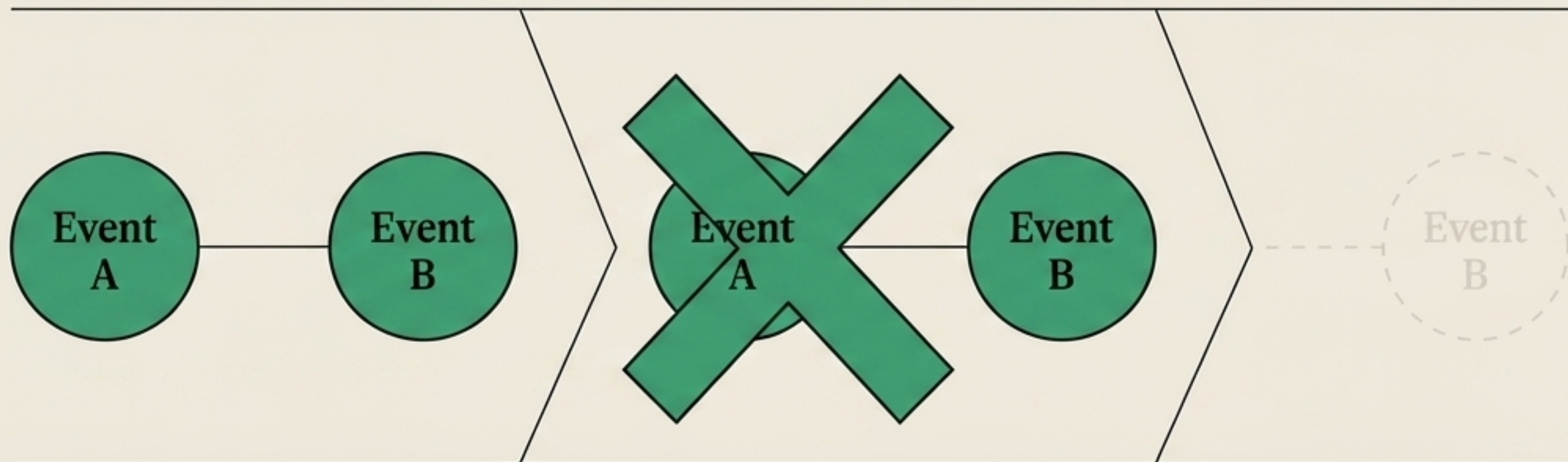
$$\begin{aligned}\text{CSCT} = & w_1 \cdot \text{Jaccard}(\text{surprise}) \\ & + w_2 \cdot \text{Gaussian}(\text{lag}, \mu, \sigma) \\ & + w_3 \cdot \text{hierarchy_match} \\ & + w_4 \cdot \text{cosine}(\text{embeddings})\end{aligned}$$



Computationally connects anomalies across all three layers.

Phase 2.5: Causal Proof — CAP

Correlation is a guess. Computational Ablation is proof.



Correlation: Friday traffic creates both. Cause or coincidence?

Computational Ablation: Physically remove data and recompute.

Result: Event B completely vanishes.

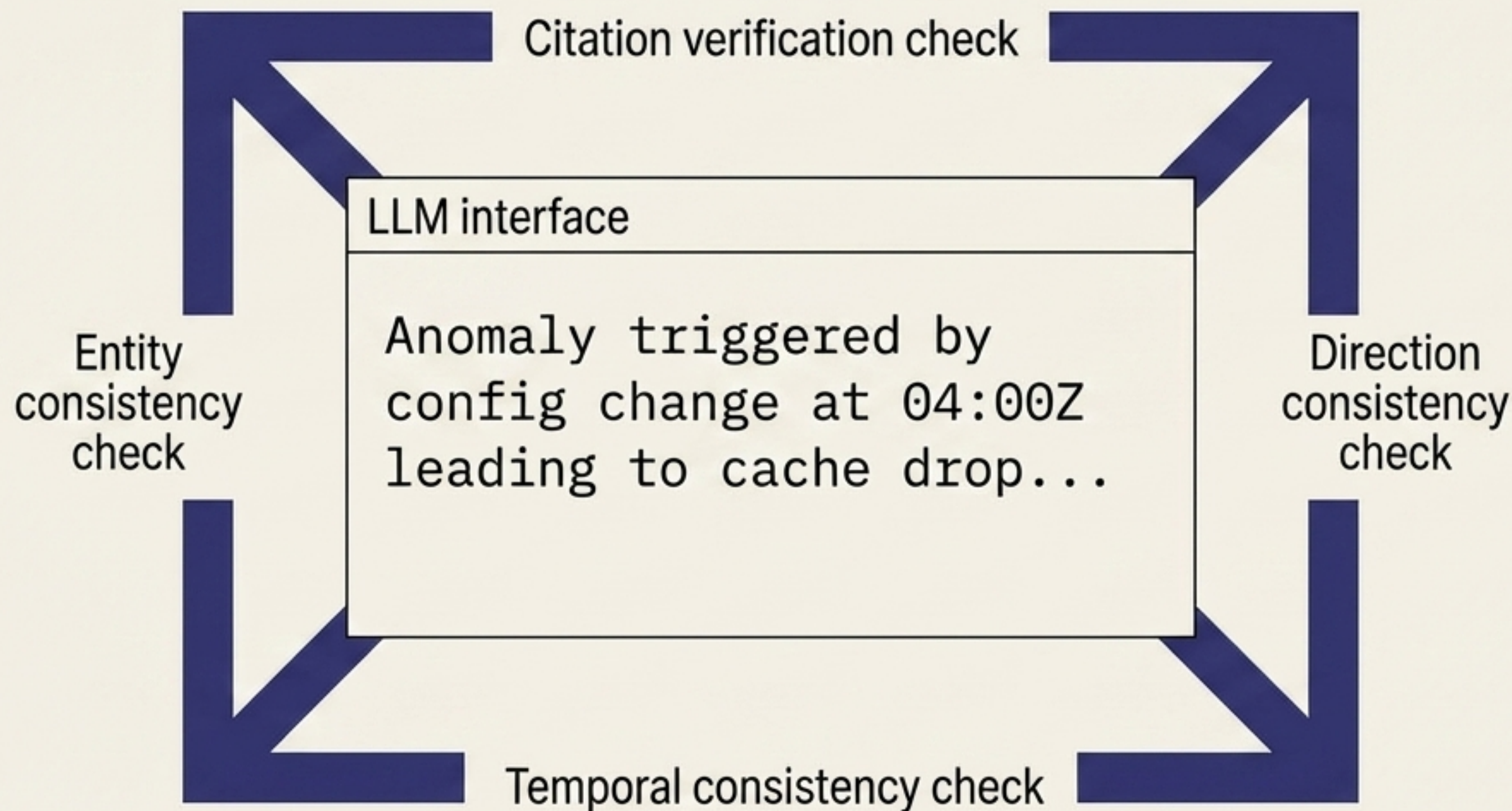
3-test causal proof methodology:

1. Remove A's users → does B vanish?
2. From A alone → can we predict B?
3. Is every link in the chain needed?

Proof Standard:
1,000× bootstrap,
95% Confidence Interval

Proven causation. Not correlation. Not guessing.

Phase 3: The Final Representation — CIL



Constrained LLM with statistical veto.

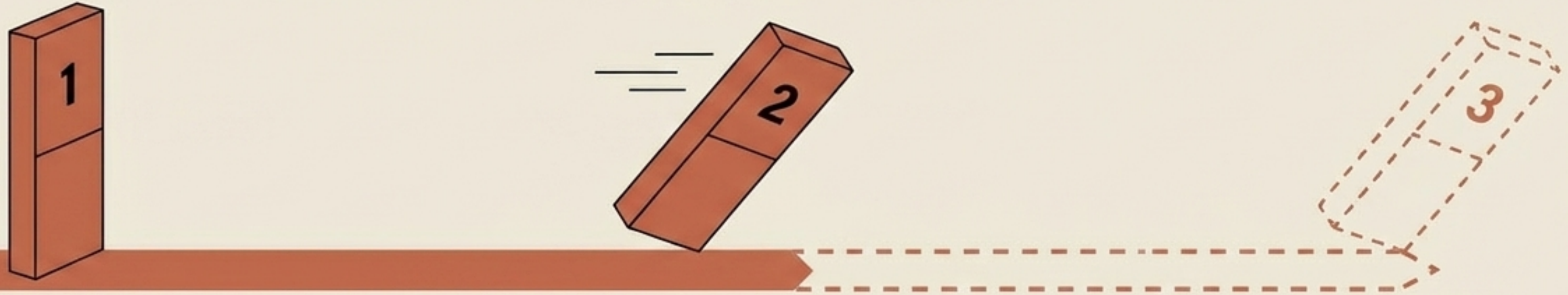
The system explains why it broke and retrieves the exact postmortem from the last time it happened.

But the math has **VETO POWER**. If the LLM narrative contradicts the proven consistency the proven causal graph from CAP, the hypothesis is silently killed.

RAG retrieval → LLM
hypothesis → citation
verification → direction check
→ temporal & entity check

Any fail = hypothesis REJECTED

The Future: Predictive Precursors — PPE



Cross-stratum precursor matching. The system learns temporal sequences from past causal graphs. When the first dominoes fall, it warns the team hours before impact.

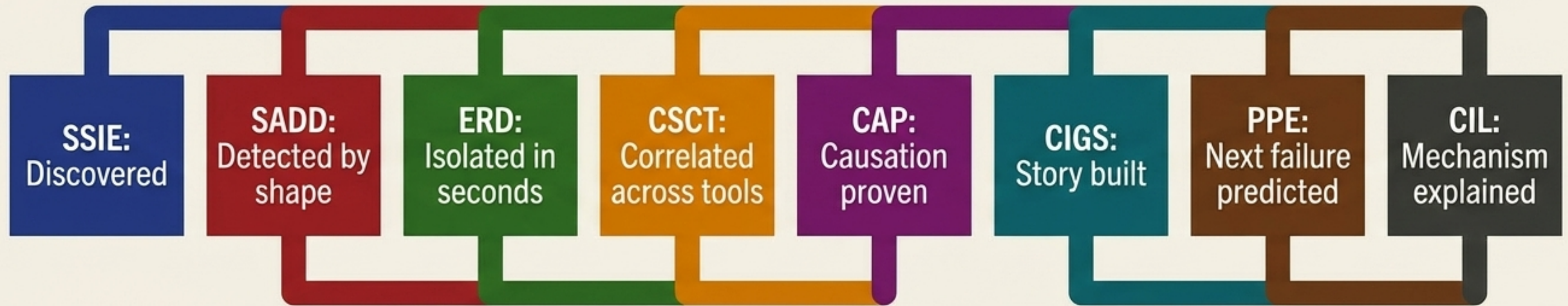
- **Anti-correlation:** Detects MISSING expected responses—finding hidden bugs by flagging the ABSENCE of data.

**A 3-hour early warning system.
Reactive triage eliminated.**

$$P(\text{incident} \mid k \text{ of } n \text{ matched}) \\ = \text{completion_rate} \times e^{(-\lambda \times \text{age})}$$

The Glass Box Architecture

You no longer have dashboards filled with guessing.
You have an automated pipeline of proof.



Six fatal flaws eliminated. A paradigm shift from thresholds to topology, from correlation to computational truth.